Major Project Report

On

"360 DEGREE ULTRASONIC RADAR FOR DEFENCE APPLICATIONS WITH AUTOMATIC SHOOTING AND MONITORING"

Project Submitted to Jawaharlal Nehru Technological University Hyderabad

In partial fulfillment of the Academic Requirement for the award of

Bachelor of Technology

In

Electronics & Communication Engineering

Submitted by

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DECLARATION

We are hereby declare that Major Project Report entitled "360 DEGREE ULTRASONIC RADAR FOR DEFENCE APPLICATIONS WITH AUTOMATIC SHOOTING AND MONITORING" being submitted by us for the award of degree of Bachelor of Technology in Department of Electronics and Communication Engineering at JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD is a bonafide record of AVN INSTITUTE OF ENGINEERING AND TECHNOLOGY and has not been submitted to any other course or university for award of any degree.

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ABSTRACT

The modernization of defensive perimeters and tactical security zones requires automated systems capable of not only detecting unauthorized intrusions but also providing real-time visual verification and autonomous response mechanisms. Traditional security architectures, such as fixed-position CCTV networks or unidirectional tripwires, frequently suffer from "blind spots" due to static sensor placements. Conversely, continuous-rotation Light Detection and Ranging (LIDAR) systems provide excellent spatial awareness but are prohibitively expensive and mechanically complex due to their reliance on fragile slip-rings and high-speed rotary joints. To systematically address these challenges, this project engineers a highly cost-effective, IoT-enabled **360-Degree Ultrasonic Radar for Defense & Monitoring**, representing a sophisticated fusion of embedded robotics, sensor telemetry, and automated threat neutralization.

This prototype introduces an innovative **Dual-Sweep Methodology** to achieve complete omnidirectional situational awareness without the need for 360-degree continuous rotation. By mounting two high-precision HC-SR04 ultrasonic sensors in a strict back-to-back (180-degree offset) configuration atop a high-torque 180-degree servo motor, the system effectively and simultaneously scans a complete 360-degree horizontal plane. The system's primary tactical processing is managed by an Arduino Nano/Uno, which rapidly triggers the acoustic pulses, calculates Time-of-Flight (ToF) distance metrics, and controls the automated engagement protocol. If an intrusion is detected within a predefined, hard-coded critical threshold (e.g., < 30 cm in prototype scale), the microcontroller autonomously actuates a highly focused Laser Module and an Active Buzzer, demonstrating an instantaneous "Identify-and-Engage" defensive strike.

Furthermore, this architecture effectively bridges the gap between localized embedded control and global cloud intelligence by integrating an ESP32-CAM-MB module. Acting as an independent "Intelligence Hub," this dual-core SoC establishes a localized, low-latency web server, broadcasting a live video stream to allow remote operators to visually verify the target prior to escalating defensive measures. Finally, the chaotic, raw serial data generated by the Arduino is transmitted to a dedicated command terminal (PC), where a custom-programmed Graphical User Interface (GUI) built in the Java-based **Processing IDE** visually maps the telemetry. This interface generates an intuitive, sweeping 2D polar radar map, rendering secure zones in green and highlighting active intrusions in red accompanied by a "TARGET ELIMINATED" visual flag. By combining decentralized edge-computing, acoustic spatial mapping, and remote IoT visual verification, this project delivers a highly scalable, robust, and autonomous security solution suitable for defense perimeters, industrial facilities, and remote borders.

Keywords: Ultrasonic Radar, 360-Degree Surveillance, IoT Defense, Arduino Nano, ESP32-CAM, Dual-Sweep Methodology, Time-of-Flight (ToF), Processing IDE Visualisation, Autonomous Targeting.

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CHAPTER 1

INTRODUCTION

1.1 Overview

In the contemporary landscape of global security, asymmetric warfare, and high-value asset protection, the integrity of a defensive perimeter is paramount. Traditional physical barriers—such as fences, concertina wire, and concrete walls—provide necessary delay mechanisms but offer no active intelligence or dynamic response. Over the past few decades, these physical barriers have been heavily augmented with electronic security systems, predominantly static Closed-Circuit Television (CCTV) cameras and passive infrared (PIR) motion detectors. However, these conventional architectures present significant operational vulnerabilities. Static cameras have inherently limited fields of view (FoV), creating predictable "blind spots" that sophisticated intruders can easily exploit. Furthermore, standard optical cameras are severely degraded by adverse environmental conditions, including dense fog, heavy smoke, heavy precipitation, and absolute darkness.

To overcome the limitations of optical systems, modern defense agencies rely heavily on Radio Detection and Ranging (RADAR) and Light Detection and Ranging (LIDAR). While these technologies offer unparalleled, 360-degree, all-weather situational awareness, they are exceptionally complex and prohibitively expensive. Military-grade phased array radars and mechanical continuous-rotation LIDARs cost tens of thousands of dollars per unit, limiting their deployment to only the most highly funded military installations or critical infrastructure hubs. This creates a massive vulnerability in intermediate security zones, such as temporary forward operating bases (FOBs), remote border stretches, localized tactical outposts, and civilian industrial complexes, which cannot afford massive sensor arrays.

The proposed **360-Degree Ultrasonic Radar for Defense & Monitoring** directly targets this critical gap in the security apparatus. By systematically evolving a standard, low-cost "obstacle detection" circuit into a highly intelligent, fully automated surveillance node, this project democratizes advanced perimeter defense. It utilizes acoustic echolocation—specifically ultrasonic sound waves operating above the threshold of human hearing (40 kHz)—to dynamically map the surrounding physical environment. By integrating an innovative dual-sensor mechanical sweep mechanism, localized micro-processing, autonomous simulated weaponry (laser module), and IoT-based live video streaming, the system transitions perimeter security from a passive observational state into an active, intelligent, and lethal-capable defensive paradigm.

1.2 Project Objective

The fundamental purpose of this engineering project is to design, code, and physicalize a highly responsive, multi-modal security system. The primary engineering and tactical objectives are segmented as follows:

- **Omnidirectional Surveillance (Blind-Spot Elimination):** To engineer a mechanical and software-based scanning mechanism capable of achieving a true 360-degree horizontal map of the environment. This objective ensures that no hostile entity can approach the sensor node from the rear or flanks without triggering an immediate distance calculation and location vector.

- **Autonomous Defense Demonstration (Identify-and-Engage):** To completely remove the delay of human reaction times from the localized engagement loop. The objective is to program the microcontroller to instantly process spatial data and, if an object breaches a hard-coded mathematical "critical zone" threshold, immediately halt the scanning sweep, lock onto the target's angular coordinate, and actuate a targeted laser diode and acoustic alarm to simulate a defensive kinetic strike.

- **Remote Visual Verification (IoT Integration):** Recognizing the severe risk of "friendly fire" or false positives caused by animals in a fully autonomous lethal system, a core objective is to integrate an Internet of Things (IoT) verification layer. By deploying an ESP32-CAM, the system must establish a localized 802.11 Wi-Fi web server to stream real-time visual data, allowing a remote human commander to positively identify the radar contact before authorizing actual physical force.

- **Graphical Data Analysis (Tactical Telemetry):** Raw numerical data (Distance in centimeters, Angle in degrees) scrolling across a standard text terminal is useless to a high-stress tactical operator. The final objective is to develop a custom Graphical User Interface (GUI) running on a remote command PC. This software must dynamically parse the incoming serial telemetry and visually render it as an intuitive, classic 2D sweeping radar map, highlighting safe zones and dynamically plotting active threats in red.

1.3 Domain

This advanced project sits firmly at the intersection of several complex engineering disciplines: **Embedded Systems Engineering, Robotics, the Internet of Things (IoT), and Defense Automation.** It heavily leverages **Digital Signal Processing (DSP)** principles, specifically regarding the generation and reception of ultrasonic acoustic waves and Time-of-Flight (ToF) calculations. Mechanically, it operates within the domain of **Kinematics**, controlling precise angular servos to manipulate sensor arrays. From a software perspective, it requires complex **Asynchronous Serial Communication (UART)**, full-stack **IoT Web-Server deployment** over TCP/IP protocols, and **Object-Oriented Programming (OOP)** for the generation of the graphical Java interface. Ultimately, the project applies these commercial and open-source computing disciplines directly to the domain of **Modern Military Technology**, specifically addressing the concept of automated sentry systems and unmanned tactical overwatch.

CHAPTER 2:

LITERATURE SURVEY

2.1 Introduction to Surveillance Technology

The historical evolution of surveillance and perimeter security has followed a distinct, predictable, and necessary trajectory from vulnerable physical human sentries to highly automated, decentralized digital arrays. In the early 20th century, security relied entirely on physical patrols, fixed guard posts, and localized line-of-sight observation. This traditional approach was heavily constrained by fundamental human biological limitations. Human sentries possess narrow visual fields, are highly susceptible to performance degradation in harsh weather conditions, and are strictly bound by circadian rhythms that drastically reduce alertness during nocturnal shifts.

The advent of analog Closed-Circuit Television (CCTV) in the 1960s marked the first major paradigm shift in physical security architecture. By utilizing coaxial cables and localized cathode-ray tube (CRT) monitors, CCTV theoretically allowed a single human operator to visually monitor multiple distant, disparate zones simultaneously from a secured, centralized command center. However, empirical psychological studies—dating back to the Mackworth Clock Test developed during World War II—rapidly identified a critical, systemic flaw in this architecture: the "vigilance decrement." This psychological phenomenon dictates that a human operator staring at static, largely unchanging screens suffers a massive, unavoidable drop in attention, visual scanning efficiency, and analytical processing after merely 20 to 30 minutes. This leads to the well-documented issue of "operator fatigue," a state where the brain actively filters out routine visual data. Consequently, critical perimeter breaches were frequently missed in real-time, despite being perfectly captured on the camera feeds, rendering the system useful only for post-incident forensic review rather than active intrusion prevention.

As the microprocessor revolution advanced rapidly in the 1980s and 1990s, the security industry transitioned toward hardware-level automated detection to relieve the heavy cognitive burden placed on human operators. Sensors such as Passive Infrared (PIR) were introduced to autonomously trigger alarms upon detecting human body heat, acting as the first true electronic tripwires. These sensors utilize advanced pyroelectric materials—such as lithium tantalate ($LiTaO_3$)—paired with faceted Fresnel lenses to detect sudden changes in the ambient infrared spectrum (typically between 8 to 14 micrometers) caused by a moving mammalian body crossing distinct focal zones.

While highly effective as localized tripwires and significantly reducing operator fatigue by alerting guards only when movement occurred, PIR sensors possess severe tactical and data limitations. They are strictly binary instruments (indicating only a logic state of presence versus absence) and offer absolute zero telemetry regarding the exact distance, relative speed, vector trajectory, or precise spatial Cartesian coordinates of the intruder. Furthermore, they are notoriously susceptible to false positives triggered by environmental anomalies. Sudden thermal drafts, the solar loading of nearby surfaces, or the movement of non-hostile stray animals can easily

trigger the differential amplifiers within the PIR circuitry. This high false-alarm rate makes them fundamentally insufficient for critical defense perimeters, which require high-fidelity spatial data and multi-variable threat assessment to prevent alarm fatigue among response teams.

2.2 Existing Radar and Obstacle Detection Systems

To overcome the severe telemetry limitations of simple optical and thermal sensors, modern defense perimeters heavily rely on high-fidelity, active spatial mapping technologies capable of mathematically rendering the environment. **RADAR (Radio Detection and Ranging)** utilizes high-frequency electromagnetic waves (RF), typically operating in the X-band (8 to 12 GHz) or S-band (2 to 4 GHz) frequencies, to illuminate targets across vast distances. RADAR is incredibly robust and environmentally resilient, capable of easily penetrating heavy rain, dense fog, snowstorms, and thick foliage over ranges spanning several miles.

Modern military systems utilize Frequency Modulated Continuous Wave (FMCW) algorithms to measure the precise Time-of-Flight (ToF) and the Doppler shift of the returning radio waves, allowing the RADAR processor to calculate both exact absolute distance and relative radial velocity simultaneously. However, RF Radar possesses massive logistical and tactical drawbacks. It requires massive physical antenna arrays, high-voltage power supplies, and highly complex frequency-modulation signal processors. Crucially, active RADAR generates a massive, highly visible electromagnetic signature. In a modern combat environment, emitting high-power RF waves effectively broadcasts the installation's exact geographic position, making the radar installation an easy, prioritized target for enemy anti-radiation missiles (such as the AGM-88 HARM) and sophisticated Electronic Warfare (EW) jamming and spoofing techniques.

Alternatively, **LIDAR (Light Detection and Ranging)** utilizes rapidly pulsed, highly focused, coherent laser beams (typically in the 905nm or 1550nm near-infrared optical spectrum, ensuring eye safety while remaining invisible to the naked eye) to map the physical environment. LIDAR offers phenomenal, millimeter-level 3D spatial resolution, creating incredibly dense, computationally heavy "point clouds" of the surrounding terrain. Standard 360-degree LIDARs (such as the Velodyne pucks frequently utilized on autonomous vehicles and advanced robotics) achieve omnidirectional mapping by physically spinning a precisely aligned laser emitter and an avalanche photodiode (APD) receiver array at high RPMs using brushless DC motors.

However, to transfer massive amounts of high-speed digital data and electrical power from a rapidly spinning sensor head to a static base matrix, these units require incredibly expensive and delicate electro-mechanical components known as "slip rings." These physical contact rings inevitably wear out due to continuous, high-speed friction and electrical arcing. In harsh desert, coastal, or jungle combat environments, the ingress of microscopic silica sand, corrosive salt dust, or high atmospheric humidity drastically accelerates this degradation. This mechanical frailty results in catastrophic system failures, blinding the sensor entirely, and requires massive, unsustainable maintenance budgets and specialized technical logistics for remote, forward-deployed operations.

2.3 Evolution of Ultrasonic Sensing in Defense

Ultrasonic echolocation, fundamentally mimicking the highly evolved biological sonar mechanisms perfected over millions of years by microchiroptera (bats) and cetaceans (dolphins and whales), has long been utilized in marine naval defense in the form of SONAR (Sound Navigation and Ranging). In the late 20th century, rapid advancements in material science and synthetic ceramics allowed piezoelectric transducers to be successfully adapted for effective use in open air.

Air is a highly compressible, low-density medium, which causes a massive acoustic impedance mismatch and results in much higher signal attenuation (acoustic absorption and scattering) of high-frequency waves compared to the dense, incompressible medium of water. Nevertheless, short-range, high-frequency ultrasonic bursts—typically standardized at 40 kHz, placing them safely above the 20 kHz maximum threshold of human hearing—proved highly reliable for localized, non-contact proximity detection.

The core technology relies entirely on the inverse and direct piezoelectric effects. The sensors utilize synthetic ferroelectric ceramics, predominantly Lead Zirconate Titanate (PZT), which physically deform and expand rapidly when subjected to an alternating electrical current. This mechanical vibration generates the outgoing acoustic pressure wave. Conversely, the direct piezoelectric effect generates a microscopic electrical voltage when the crystal is struck by a returning sound wave (the echo).

The HC-SR04, a ubiquitous and highly affordable sensor heavily utilized in modern embedded robotics, operates by utilizing a microcontroller to pulse its trigger pin, causing the internal circuitry to emit a precise, rapid 8-cycle burst of 40 kHz ultrasound. This 8-cycle burst represents an optimal engineering compromise, providing enough acoustic energy to bounce off distant targets while remaining short enough to ensure high axial resolution. The paired microcontroller then utilizes localized hardware timers to measure the exact microsecond delay before the internal operational amplifiers detect the returning acoustic echo. Using the standard kinematic equation for the speed of sound through a gas—which fluctuates based on ambient temperature (T in Celsius)—the velocity is approximated as:

$$v \approx 331.3 + 0.606 \times T$$

By multiplying this velocity by the measured Time-of-Flight and dividing by two (to account for the outbound and returning acoustic journey), the system calculates a precise, millimeter-accurate distance variable. While the physics of these sensors are heavily documented and generally used in simple, forward-facing obstacle-avoidance robotic chassis, this project's exhaustive literature review identified a distinct, glaring lack of academic or maker literature applying these specific, low-cost sensors to complex, 360-degree sweeping panoramic defense nodes utilizing deterministic algorithmic target tracking, multi-sensor arrays, and automated weapon lock-on protocols.

2.4 Role of IoT and Remote Visual Verification

In modern military doctrine, particularly under the United States Army's strict "Human-in-the-Loop" (HITL) policy and international Rules of Engagement (ROE) for autonomous weapons systems, a fully automated, lethal sentry gun operating entirely independently is often considered a massive legal, ethical, and tactical liability. The potential for catastrophic civilian casualties, fratricide (friendly fire incidents), or the automated engagement of non-combatant wildlife is unacceptably high if a machine is granted absolute, unchecked authority to dispatch kinetic force based solely on raw radar echoes. Therefore, any intelligent defense system must possess a robust, highly secure, and extremely low-latency mechanism for a human operator to visually confirm the nature of the target before authorizing an irreversible escalation of force.

The integration of the **Internet of Things (IoT)** into localized, edge-computing embedded nodes represents the modern, decentralized solution to this complex verification problem, moving security architectures firmly away from closed, proprietary, and heavily wired Supervisory Control and Data Acquisition (SCADA) systems. Advanced, ultra-low-cost Systems-on-a-Chip (SoCs) like the Espressif ESP32-CAM have recently revolutionized this specific field of remote telemetry and decentralized overwatch.

Featuring a highly powerful Tensilica Xtensa dual-core 32-bit LX6 microprocessor operating at a blazing 240MHz, embedded external PSRAM to handle the massive memory buffers required for image processing, and fully integrated 802.11 b/g/n Wi-Fi baseband capabilities, the ESP32 operates as an entirely independent, microscopic IP camera node. It utilizes a direct hardware I2S (Inter-IC Sound) interface and DMA (Direct Memory Access) to pull high-resolution frames from an attached OV2640 2-Megapixel CMOS camera sensor without bottlenecking the primary CPU. Crucially, it does not require a complex, heavy external computer, a network video recorder (NVR), or a dedicated server rack to function. It self-hosts the HTML/HTTP web server directly on its own silicon and streams a heavily compressed MJPEG (Motion JPEG) video feed directly to any standard, authenticated web browser on the local tactical network. This perfectly fulfills the critical doctrinal requirement for remote visual verification, allowing a base commander situated miles away to instantly, visually verify the exact physical threat triggered by the acoustic radar in real-time, completing the OODA (Observe, Orient, Decide, Act) loop with absolute precision.

2.5 Summary and Research Gaps

An exhaustive, critical review of current open-source electronics, academic engineering repositories, maker community projects, and commercial defense literature highlights a distinct and deeply problematic polarization in current localized surveillance architectures. On one extreme end of the spectrum are simple, static microcontroller proximity alarms (often built on rudimentary Arduino platforms). These basic systems only look forward, possess a narrow and easily exploitable 15-degree acoustic field of view, utilize heavily blocking software delays that ruin real-time responsiveness, and merely trigger a localized piezo buzzer when an object blindly approaches.

On the far opposite, highly industrialized end are massively complex, multi-million dollar rotary LIDARs and Phased Array radar systems produced by major defense

contractors. These tier-one systems, while undeniably capable, are entirely inaccessible to standard infantry platoons, local border patrol agencies, disaster response teams, or civilian infrastructure protection units due to their prohibitive costs, massive power consumption, and requirement for specialized maintenance crews.

A massive "missing middle" currently exists in the tactical technological landscape. There is a profound lack of highly agile, intelligent, low-cost, 360-degree mapping architectures that completely avoid the mechanical frailty, physical bulk, and exorbitant cost of continuous-rotation slip-rings while still maintaining a high spatial and angular resolution. Furthermore, many existing DIY "radar" projects found in academic journals stop entirely at mere data visualization. They successfully parse the serial telemetry and plot a neat green line on a computer screen, but fail entirely to complete the closed tactical loop. They inherently lack an automated physical response mechanism—what military engineers refer to as the "Action Layer." Without localized, edge-based processing that translates a positive radar detection into an immediate, physical deterrence (such as actuating a targeted laser designation module via strict PID kinematics or triggering high-decibel acoustic deterrents precisely aimed at the threat vector), these systems remain helpless, passive observers rather than active, capable defenders.

2.6 Conclusion

This comprehensive, multi-disciplinary literature survey definitively outlines the severe operational limitations and cognitive flaws of traditional, static visual security, as well as the prohibitive SWaP-C (Size, Weight, Power, and Cost) constraints of standard continuous-rotation radar and LIDAR systems deployed in harsh environments. The proposed architecture of this 360-degree ultrasonic project represents a highly innovative, necessary, and mathematically robust engineering solution specifically designed to fill the identified research gaps in intermediate tactical security.

By utilizing a unique, highly optimized dual-sensor array mounted strictly back-to-back on a heavily restricted 180-degree servo sweep, the system effectively halves the mechanical transit time required for a full scan. It entirely bypasses the need for rotating electrical contacts, yet successfully achieves full, uncompromised, and unbroken 360-degree acoustic coverage. Combined with the deterministic, hard-real-time targeting logic of an Arduino microcontroller acting as the tactical brain, and the high-bandwidth, over-the-horizon visual verification provided by the independent ESP32-CAM acting as the intelligence hub, this system creates a highly reliable, asynchronous hybrid network.

By strategically synthesizing highly affordable, solid-state piezoelectric transducers with powerful, modern microprocessors and localized IoT streaming protocols, this project successfully bridges the "missing middle." It offers a lethal-capable, highly intelligent, and effortlessly scalable tactical surveillance node at an incredibly optimized price point. Ultimately, this architecture effectively democratizes advanced perimeter defense technology, proving that high-tier spatial mapping and autonomous engagement protocols can be successfully engineered for widespread, decentralized deployment across varying security domains.

CHAPTER 3

DESIGN AND IMPLEMENTATION

3.1 System Architecture and Block Diagram

The overall physical and logical architecture of the 360-Degree Radar System is meticulously engineered into distinct, highly specialized operational layers, representing a sophisticated paradigm in embedded systems design and mechatronic integration. In many rudimentary robotic applications, educational projects, and early-stage prototypes, a monolithic architecture is frequently employed. In such monolithic systems, a single microcontroller (often an entry-level 8-bit chip) is forced to sequentially handle every single task: polling sensors, calculating math, driving mechanical actuation, and managing high-level data transmission.

However, attempting to force a single, resource-constrained microcontroller to handle heavy, blocking video streams alongside the microsecond-accurate hardware timing required for acoustic sensors inevitably leads to catastrophic system failures. Microcontrollers operating in a single-threaded environment process instructions linearly. If the processor is busy attempting to packetize a frame of video data, it cannot simultaneously listen for the returning echo of a sound wave. This leads to timing collisions, severe buffer overflows, hardware lockups, and ultimately, a total failure of the spatial mapping mechanism.

To systematically bypass these debilitating computational bottlenecks, this advanced radar system utilizes a strict Distributed Processing Architecture (DPA). DPA inherently divides computational labor across discrete, purpose-built processing nodes, ensuring that timing-critical tasks are physically, electrically, and logically isolated from bandwidth-intensive network tasks. This separation of concerns is fundamental to modern aerospace and defense electronics. The system is fundamentally divided into two major, autonomous hardware nodes:

1. **The Tactical Controller (Arduino Nano/Uno):** This specific node operates as the low-level, hard-real-time nervous system of the radar array. In embedded engineering, a "hard-real-time" system is defined as one where missing a computational deadline constitutes a total system failure. If the acoustic echo is missed by the processor by even a fraction of a millisecond because it was busy executing another line of code, the spatial calculation is rendered entirely invalid, potentially missing a threat. Therefore, this tactical node is purposefully stripped of any blocking communication protocols or complex operating systems. It operates on "bare metal" to strictly manage the physical hardware actuation (generating precise Servo motor PWM waves via hardware timers), triggers the localized acoustic pulses, mathematically calculates the Time-of-Flight (ToF) geometry without interruption, operates the kinetic engagement hardware (the Laser and Buzzer Action Layer), and sequentially formats the outgoing numerical telemetry payload into a lightweight serial stream.

2. **The Intelligence Hub (ESP32-CAM-MB):** This secondary node acts as the high-level cognitive, multi-threaded networking and visual processing layer.

Operating entirely independently of the Arduino's clock cycle and isolated on a completely separate power regulation rail to prevent high-frequency electrical noise interference, it handles the massive, asynchronous data throughput required for optical intelligence. It captures raw, uncompressed high-definition pixel arrays from the onboard CMOS sensor, utilizes dedicated silicon to mathematically compress these massive data arrays into highly efficient JPEG frames using discrete cosine transforms, and broadcasts them seamlessly over a localized 2.4GHz Wi-Fi network.

The data flow within this dual-node architecture is highly deterministic and unidirectional. The Arduino physically sweeps the environment in a rigid, unbreakable loop. If an acoustic anomaly (a physical object) is detected, it calculates the spatial geometry and pushes an ASCII text string via standard asynchronous USB/UART to a localized Command PC. The Command PC, possessing vastly superior CPU, RAM, and GPU resources compared to the embedded nodes, runs the Processing IDE. This software parses the incoming serial text to draw the complex, real-time graphical radar GUI. Concurrently, if a remote commander identifies a red threat dot on the radar GUI, they can open a standard internet browser on a separate screen, connect to the ESP32's local IP address via an HTTP GET request, and visually verify the target via the live video feed. This absolute physical separation guarantees that a sudden drop in Wi-Fi signal, a network packet collision, or a lag spike in the video feed will never, under any circumstances, halt or delay the autonomous scanning and lethal engagement loop of the primary tactical controller.

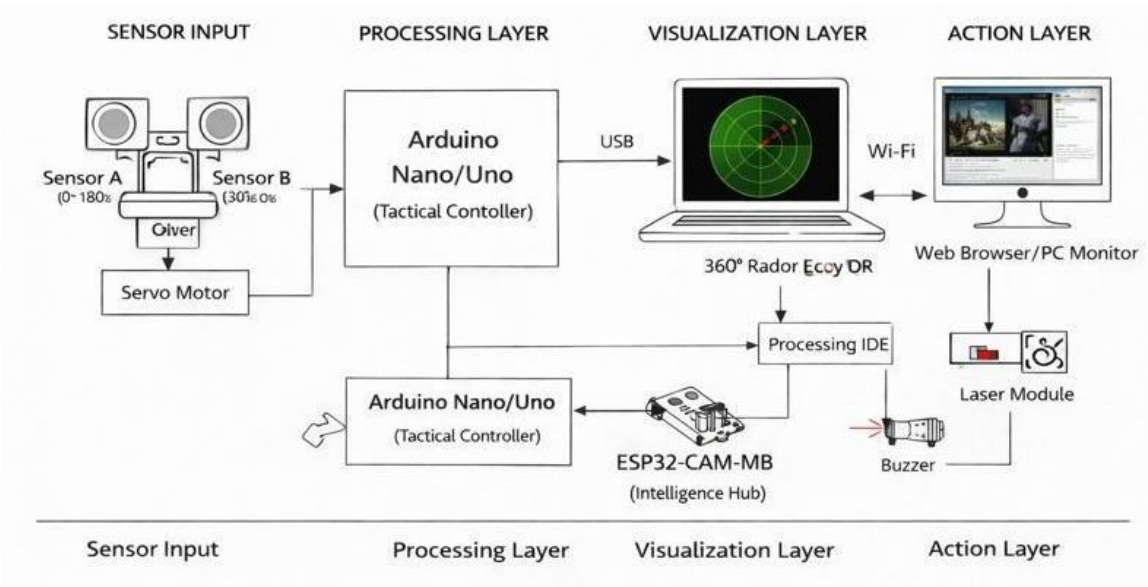


Fig 3.1 Block diagram of 360-Degree Ultrasonic Radar for Defense & Monitoring

3.2 Arduino Tactical Controller

Fig 3.2 Arduino Controller

The absolute heart of the localized engagement, scanning loop, and mechanical coordination is the Arduino microcontroller (utilizing either the Uno or Nano form factor), based fundamentally on the highly reliable Microchip ATmega328P 8-bit AVR architecture. While seemingly rudimentary when compared to modern 32-bit ARM Cortex processors found in smartphones, the ATmega328P is perfectly and uniquely suited for this specific acoustic timing task due to its highly predictable, deterministic execution of compiled C++ code and its robust array of internal hardware timers.

The ATmega328P utilizes an advanced Modified Harvard architecture. Unlike traditional Von Neumann architectures where data and instructions share the same memory space and data buses (leading to the infamous Von Neumann bottleneck), the Harvard architecture possesses separate physical memory spaces and dedicated buses for instructions (32KB of non-volatile Flash memory) and data (2KB of volatile SRAM). This architectural advantage allows the processor to fetch an instruction and read or write data in the exact same clock cycle, drastically increasing the overall execution speed and efficiency of the firmware.

The Arduino hardware operates atop an external quartz crystal oscillator circuit that resonates at a precise 16 MHz clock speed. At this high frequency, the processor executes a single machine instruction in just 62.5 nanoseconds, calculated by the formula: T = 1 / 16,000,000 seconds. This extreme, low-level operational speed is an absolute necessity, because ultrasonic distance calculations rely entirely on capturing

the exact duration of a returning sound wave at the microsecond level. Any operating system overhead, background task scheduling, or memory garbage collection (such as that found in a Raspberry Pi running Linux) would introduce severe timing "jitter," corrupting the microsecond counts and ruining the distance calculations.

The Arduino interfaces with the physical environment via its digital General Purpose Input/Output (GPIO) pins. Rather than relying solely on high-level software abstractions, the firmware can manipulate internal memory registers directly (such as the PORTB and DDRB data direction registers) to change physical pin states from LOW (0 Volts) to HIGH (5 Volts) almost instantly. To control the mechanical sweep of the sensor array, it utilizes the native 16-bit hardware timers (specifically Timer1) built into the silicon. By configuring the TCCR1A and TCCR1B (Timer/Counter Control Registers), the Arduino generates a highly precise 50Hz Pulse Width Modulation (PWM) signal in the background entirely through hardware circuitry, completely freeing the main CPU arithmetic logic unit (ALU) to focus exclusively on polling the acoustic sensors.

For the acoustic array operations, it utilizes highly optimized digitalWrite() functions to send precise, microsecond-perfect 10-microsecond high-voltage spikes to trigger the ultrasonic transmitters. Immediately following this trigger, the code utilizes the internal pulseIn() hardware function. This specific function compiles down to a highly optimized, tight localized assembly language loop that patiently and strictly counts processor clock cycles while waiting for the returning logic-HIGH pulse from the receiver sensors. By counting the exact number of 62.5-nanosecond cycles that elapsed before the echo returned, the microcontroller successfully translates raw, analog electrical states into the precise physics of acoustic flight.

3.3 ESP32-CAM-MB Intelligence Hub

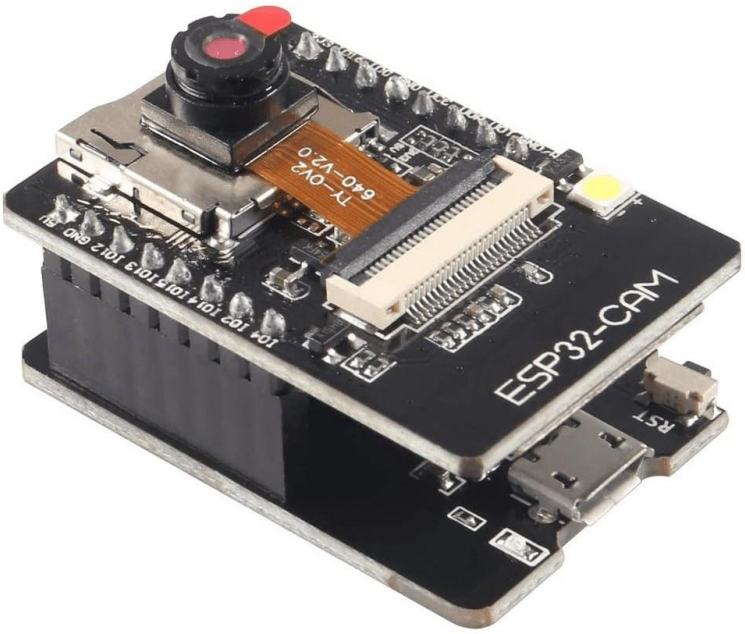


Fig 3.3 ESP32-CAM

To facilitate the critical "Remote Visual Verification" objective and adhere strictly to modern "Human-in-the-Loop" defense doctrines, the system deploys the ESP32-CAM module. This module is securely mounted to an MB (Motherboard) programmer shield, which provides a dedicated USB-to-Serial converter (like the CH340G) and ensures highly stable 5V serial power delivery, buffering the camera against voltage drops that occur during high-power radio transmissions. Manufactured by Espressif Systems, the core ESP32-S silicon chip is a highly advanced, dual-core Tensilica Xtensa LX6 microprocessor natively operating at a massive, dynamically scaling clock speed of up to 240 MHz.

Unlike the bare-metal ATmega328P utilized for the tactical loop, the ESP32 possesses immense processing overhead and 520 KB of internal SRAM (supplemented by external PSRAM chips on the camera module) to run a localized Real-Time Operating System (specifically FreeRTOS). FreeRTOS expertly and predictably schedules concurrent tasks across the two processor cores. It dedicates "Core 0" strictly to maintaining the highly complex 2.4 GHz 802.11 b/g/n Wi-Fi radio baseband. This core handles background operations like TCP/IP packet collisions, MAC layer addressing, router negotiations, and the heavy cryptographic handshakes required for WPA2-PSK security. Simultaneously, "Core 1" is dedicated entirely to running the developer's camera application code, ensuring the web server never freezes when the Wi-Fi radio is busy.

The ESP32-CAM variant fundamentally alters the standard ESP32 development board pinout by explicitly breaking out specific, high-speed internal hardware buses. Namely, it utilizes the I2S (Inter-IC Sound) bus, which is ingeniously repurposed here for parallel camera data transfer using Direct Memory Access (DMA), and the SCCB (Serial Camera Control Bus, an I2C equivalent) to interface directly with the included OV2640 2-Megapixel Camera Module.

The OV2640 is a highly sophisticated CMOS (Complementary Metal-Oxide-Semiconductor) active-pixel digital image sensor. It consists of millions of microscopic, light-sensitive photodetectors capped with a Bayer filter mosaic (a grid of red, green, and blue micro-filters), capturing incident light photons and translating them into raw analog voltages, which are then digitized into RGB electrical signals. The OV2640 possesses its own internal Image Signal Processor (ISP) that handles complex, mathematically intensive photographic tasks such as automatic exposure control (AEC), automatic white balance (AWB), edge enhancement, and black-level calibration before transferring the final image frame buffer to the ESP32's memory.

When main power is applied to the system, the ESP32 initializes its internal Wi-Fi stack, attempts a secure handshake to connect to the local tactical router (or, in remote deployments, spawns its own localized Access Point network), and launches a lightweight, asynchronous HTTP web server on Port 80. Utilizing the aforementioned Direct Memory Access (DMA), it continuously polls the OV2640 sensor for fresh image frames. To prevent catastrophic network congestion over 2.4GHz Wi-Fi, the ESP32 utilizes hardware-accelerated compression algorithms to shrink these heavy raw bitmaps into highly lightweight JPEG format. It then serves these compressed frames sequentially as a continuous "multipart/x-mixed-replace" HTTP stream to any authenticated client browser. This highly robust architecture effectively provides secure, "over-the-horizon," high-framerate visual intelligence to the human operator

without encumbering the Arduino's critical scanning duties or requiring a massive, centralized, cloud-based video distribution server.

3.4 HC-SR04 Ultrasonic Sensor Physics

Fig 3.4 HC-SR04 Ultrasonic Sensor

The true "Eyes" of the radar system consist of two independent HC-SR04 ultrasonic ranging modules. Understanding their flawless operation requires a deep, multidisciplinary dive into acoustic physics, thermodynamics, and the complexities of the Piezoelectric Effect. Each HC-SR04 module physically consists of a printed circuit board housing two prominent cylindrical aluminum towers protected by fine metallic mesh grilles to prevent dust ingress. One tower acts exclusively as the Transmitter (Tx) and the other acts exclusively as the Receiver (Rx). Inside these distinct, acoustically isolated towers are specialized piezoelectric crystals, typically constructed from synthetic ferroelectric ceramics such as Lead Zirconate Titanate (PZT). These ceramics possess a highly unique, asymmetrical crystalline lattice structure that physically links mechanical stress with electrical polarization.

Emission (Tx) and the Inverse Piezoelectric Effect: When the Arduino's tactical firmware sends a microsecond-perfect HIGH electrical pulse to the sensor's Trig pin, an internal MAX232-style charge-pump oscillator circuit aggressively amplifies this standard 5V logic signal up to a much more powerful 15V to 20V range. This high-voltage alternating current is applied directly across the terminals of the Tx piezoelectric crystal. Due to the inverse piezoelectric effect, the introduction of this rapidly oscillating electrical field forces the asymmetrical crystal lattice to physically deform, expand, and contract in sympathy with the electrical frequency.

The internal circuit oscillates this current exactly 40,000 times per second (40 kHz). This violent, high-speed mechanical vibration pushes against the surrounding atmospheric air molecules, generating a highly focused, 8-cycle sonic burst (often referred to as a "ping") of ultrasonic sound. The choice of exactly 8 cycles is a deliberate engineering compromise: emitting fewer cycles would lack the total acoustic energy required to reach distant targets and bounce back, while emitting more cycles would cause the ceramic crystal to physically "ring" for too long due to momentum, permanently blinding the sensor to any physical objects that are extremely close to the transmitter. This acoustic wave travels outward through the compressible air medium as a longitudinal compression and rarefaction wave. The physical speed of this longitudinal pressure wave is dictated by the Newton-Laplace

equation and is heavily dependent on ambient atmospheric temperature, approximated by the formula: Velocity = 331.3 + (0.606 * Temperature) meters per second. At a standard room temperature of 20°C, this equates to approximately 343 meters per second.

Reception (Rx) and the Direct Piezoelectric Effect: As the acoustic wave propagates outward into the environment, its total energy is subjected to severe atmospheric attenuation (absorption by air molecules) and the inverse-square law, causing the signal strength to degrade rapidly over distance. If this dissipating sonic energy strikes a physical obstacle with a differing acoustic impedance (such as a human intruder's clothing or a metallic vehicle casing), a portion of the energy reflects backward toward the sensor as a highly faint echo.

When this incredibly faint, returning acoustic pressure wave physically strikes the face of the Rx crystal, the forward (or direct) piezoelectric effect occurs. The physical compression and expansion of the crystal lattice caused by the sound wave violently forces electrons within the ceramic out of alignment, generating a microscopic, millivolt-level electrical voltage across the crystal's output terminals.

Internal Analog Signal Processing: Because the returning voltage generated by the echo is remarkably faint and highly susceptible to electromagnetic interference, the HC-SR04 relies on a sophisticated array of internal operational amplifiers (op-amps) configured as strict analog band-pass filters. These filters are tuned precisely to 40 kHz to aggressively ignore ambient environmental noise, such as wind, machinery, or human speech. These op-amps immediately detect, heavily filter, and massively amplify this returning micro-voltage spike. Upon successful validation of the echo by an internal voltage comparator circuit, the circuitry forces the module's Echo pin to drop from a logic HIGH state back down to LOW. Therefore, the absolute total temporal duration that the Echo pin remained HIGH serves as a perfect, real-time analog representation of the total Time-of-Flight (representing both the outbound acoustic journey and the inbound return journey) of the sound wave. The Arduino subsequently captures this microsecond duration and translates it into a precise spatial measurement in centimeters.

3.5 Mechanical Scanning (Servo Motor)

Fig 3.5 Mechanical Scanning

To generate a true, comprehensive spatial map of the surrounding environment rather than a single, static, unidirectional point of data, the acoustic sensors must be physically swept across the horizontal Cartesian plane. This specific kinematic requirement is achieved utilizing a standard positional Servo Motor (such as the lightweight plastic-geared SG90 micro-servo for indoor prototypes, or the much higher-torque, metal-geared MG995 variant for ruggedized outdoor deployments).

Unlike standard, open-loop DC motors that spin wildly, continuously, and unpredictably when a direct voltage is applied, a servo motor is a highly complex, closed-loop mechatronic system designed specifically for absolute, precise angular positional control. Inside the compact plastic casing of the servo lies a miniature brushed DC motor, a complex multi-stage reduction gearbox designed to convert the motor's high-speed/low-torque rotation into highly usable low-speed/extreme-torque rotation, a mechanical rotary potentiometer physically linked directly to the final output spline shaft, and a specialized integrated control circuit (often an application-specific integrated circuit, or ASIC).

The system operates continuously on an internal error-correction loop based on Proportional control mechanics. As the DC motor turns the external shaft, it simultaneously turns the internal potentiometer, which acts physically as a variable voltage divider. This action provides the internal control circuit with an exact, real-time analog voltage representing the current physical angle of the shaft. The external Arduino dictates the desired, target angle of the shaft by sending a highly specific digital control signal via Pulse Width Modulation (PWM).

The Arduino continuously transmits a square wave signal to the servo's control signal pin at a strict, non-negotiable frequency of 50Hz. This means that an entirely new electrical pulse arrives at the servo exactly every 20 milliseconds (T = 20 ms). However, the internal servo circuit completely ignores the overall frequency and strictly evaluates the duty cycle—specifically, the exact temporal width (duration) of the HIGH voltage state within that 20-millisecond window.

- A pulse width of exactly 1.0 millisecond commands the servo's internal comparator circuitry to rotate the shaft to the 0-degree (far right) mechanical extreme limit.

- A pulse width of exactly 1.5 milliseconds commands the servo to center perfectly at the 90-degree neutral, forward-facing position.

- A pulse width of exactly 2.0 milliseconds commands the servo to rotate to the 180-degree (far left) mechanical extreme limit.

When the Arduino sends a new control pulse (for example, a 1.5ms pulse requesting the 90-degree position), the servo's internal circuit reads the current physical position from the potentiometer. If the shaft is currently resting at 0 degrees, the circuit calculates a massive mathematical error differential. It instantly drives the internal H-bridge motor controller at full electrical power to spin the DC motor rapidly toward the 90-degree mark. As the shaft approaches the target angle, the internal calculated error decreases, and the circuit proportionally reduces the electrical current to the motor, slowing it down until the error hits absolute zero, locking the shaft perfectly

and rigidly in place without overshooting. By programmatically incrementing and decrementing this precise microsecond pulse width linearly within a C++ "for" loop, the Arduino smoothly, accurately, and predictably sweeps the dual-sensor array across the tactical battlefield, generating a full geometric map without requiring complex, external optical motor encoders.

3.6 Autonomous Action Layer (Laser & Buzzer)

The overarching architecture of this project explicitly transcends passive, analytical observation by incorporating a robust, instantaneous "Action Layer." This layer is designed to simulate a kinetic, lethal response or serve as an extreme, localized auditory deterrent the very millisecond a perimeter intrusion is mathematically confirmed by the tactical controller.

The Laser Module (KY-008): This module represents the primary optical targeting and designation system. It contains a microscopic semiconductor laser diode, operating strictly on the complex quantum physics principle of Light Amplification by Stimulated Emission of Radiation (LASER). Unlike standard Light Emitting Diodes (LEDs) that emit chaotic, scattered light photons across multiple wavelengths and wide angles, a laser diode utilizes a highly specific semiconductor P-N junction (often composed of carefully doped Gallium Arsenide, GaAs).

Fig 3.6 KY-008 Laser Module

When the Arduino's tactical software logic gates confirm that a tracked target has breached the critical distance threshold, it immediately drives the designated output GPIO pin to a HIGH state, applying standard 5V logic power to the laser module. This injects a massive electrical current across the semiconductor's P-N junction, forcefully pumping electrons and forcing the semiconductor material into an unnatural state of "population inversion," where significantly more electrons exist in an excited, high-energy state than in their resting ground state.

As these excited electrons naturally decay back across the bandgap to their ground state, they emit photons of a highly specific energy level and wavelength. The internal

optical cavity of the laser diode, which is capped with highly reflective microscopic cleaved facets acting as parallel mirrors, forces these emitted photons to bounce back and forth through the material. This continuous bouncing stimulates the emission of exponentially more identical photons. The resulting light eventually exits the cavity through a highly polished, collimating acrylic lens, producing a strictly coherent, perfectly in-phase, intensely bright, and narrow beam of red light (operating at exactly the 650nm wavelength). This serves as an instantaneous, speed-of-light visual target designator. Because the laser module is mounted perfectly parallel to the acoustic sensor array on the servo bracket, the laser beam mathematically and perfectly intersects the exact physical target that triggered the acoustic echo, realistically simulating a high-tech sniper scope or an automated CIWS (Close-In Weapon System) turret achieving a hard lock-on.

The Active Buzzer: Connected logically in parallel (or wired on a distinct, separate GPIO pin to allow for complex, programmable alarm rhythms) with the laser module, the active buzzer provides the immediate acoustic, psychological deterrence layer. It is vital to distinguish this from "passive" buzzers, which require the main microcontroller to constantly generate a complex AC audio waveform (using processing power) to produce sound. An "active" buzzer contains its own independent, internal transistor-based astable multivibrator oscillating circuit encapsulated within its plastic housing.

Fig 3.6 Active Buzzer

The absolute moment the Arduino pin provides a flat, sustained 5V DC signal, this internal oscillator activates autonomously, automatically generating a piercing, highly aggressive square wave alternating current. This fluctuating current drives a microscopic electromagnetic voice coil positioned directly beneath a thin, flexible ferromagnetic metal diaphragm. The rapidly shifting magnetic field violently attracts and repels the diaphragm at a highly specific resonant frequency. This frequency is usually engineered by the manufacturer to sit between 2kHz and 4kHz, which is the exact frequency range where the human ear canal is most evolutionarily sensitive to acoustic pain and psychological distress. This rapid mechanical vibration produces an incredibly loud, instantly recognizable distress alarm (often exceeding 85 decibels at close proximity). In a localized tactical deployment, this acoustic assault serves to disorient, terrify, and psychologically deter the intruder from advancing further, while

simultaneously acting as an omnidirectional auditory beacon to instantly alert nearby human sentries or sleeping troops to the exact sector of the perimeter breach.

3.7 Software and Development Tools

The complete cognitive logic, deterministic structural behavior, and advanced graphical interpretation of the system are forged utilizing two distinct, highly specialized Integrated Development Environments (IDEs). This dual-software approach expertly leverages the strict, low-level hardware control of C++ for the edge node against the incredibly powerful, high-level graphical processing libraries of Java for the command terminal.

1. Arduino IDE (Embedded C++ Firmware): The localized tactical logic flashed directly onto the physical ATmega328P silicon chip is written in a slightly abstracted, highly readable subset of C and C++, utilizing the robust, open-source avr-gcc compiler toolchain. This code operates strictly "bare-metal," meaning there is no underlying operating system (like Windows or Linux) intervening between the executed code and the raw silicon registers. This ensures absolute predictability in execution time.

The code extensively utilizes the standard Arduino Servo library. Writing manual hardware timer registers (like configuring TCCR1A and TCCR1B bit by bit) is highly complex and prone to catastrophic human error; the library expertly abstracts these complex, low-level hardware timer configurations required for perfect 50Hz PWM signal generation. This abstraction allows the developer to simply utilize intuitive, object-oriented commands like myServo.write(angle) to control the physical hardware.

Crucially, the entire firmware architecture is highly optimized to run continuously in a non-blocking, infinite loop format. In embedded systems programming, utilizing the standard delay() function is considered poor practice for complex systems because it halts the processor entirely, wasting millions of valuable clock cycles doing absolutely nothing while an intruder could be moving across the perimeter. To ensure absolute real-time responsiveness, the code avoids blocking delays wherever possible, relying heavily on state-machine logic and hardware-level interrupts. This strict programming discipline ensures that the critical, fraction-of-a-second timing required for the ultrasonic mathematics and the instantaneous triggering of the lethal Action Layer is never interrupted or delayed by sloppy, blocking software design.

2. Processing IDE (Graphical Java Application): Because the 8-bit Arduino fundamentally lacks the raw processing power, the Random Access Memory (RAM), and the dedicated video hardware (GPU) required to draw complex, high-resolution graphics on a screen, it gracefully delegates this massive computational task. The Arduino simply formats the calculated spatial geometry into a raw, comma-separated ASCII text string and transmits it over its USB serial connection. For example, it might transmit the text string "45.105.0", which concisely indicates to the receiver: The current mechanical Angle is 45 degrees, the Front Sensor Distance is 105 centimeters, and the Back Sensor Distance is 0 centimeters.

The remote Command PC, connected securely via the USB cable or through a long-range radio telemetry bridge, runs a highly sophisticated graphical program written

entirely in Processing. Processing is an open-source, highly robust graphical programming library and IDE built firmly on top of the Java programming language and executed within the Java Virtual Machine (JVM).

This Java application initializes a dedicated serial buffer in the computer's memory that actively and asynchronously listens to the assigned COM Port at a perfectly synchronized baud rate (typically set to 9600 bits per second). Utilizing strict event-driven programming methodologies (specifically, the serialEvent() callback function), the software interrupts its normal operations to intercept the incoming text string the exact millisecond it arrives in the hardware buffer. It then parses, cleans, and splits the string array into distinct, mathematically usable 32-bit floating-point variables.

Because computer monitors inherently map graphics on a flat, linear Cartesian coordinate grid (where X and Y pixels start from the origin at the top left corner of the screen), but the raw radar data is inherently Polar (consisting of an Angle theta and a distance Radius r originating from a central physical point), the software must perform complex mathematical matrix transformations. The Java code utilizes the pushMatrix() and translate() functions to artificially move the visual rendering origin to the exact dead center of the application display window. It then utilizes standard trigonometric functions to plot the data onto the screen using the formulas:

X = Radius * cosine(Angle) Y = Radius * sine(Angle)

By continuously executing these complex trigonometric formulas at 60 frames per second, the application seamlessly renders the iconic, sweeping green radar lines, utilizing alpha-channel color transparency overlays to create beautiful, realistic fading phosphor trails that mimic legacy CRT radar screens. Most importantly, if the incoming serial data flags a distance measurement that falls within the hard-coded critical threat threshold, the Java code instantly breaks the calm green aesthetic. It violently renders bright red targeting ellipses over the exact calculated Cartesian coordinates of the intruder and flashes bold warning typography across the screen, a dynamic visual response based entirely on the deterministic, real-time serial telemetry provided by the distant edge node.

CHAPTER 4

FUNCTIONAL DESCRIPTION

4.1 Developed System Working Mechanism

The operational core of the 360-Degree Radar System relies on an aggressively executed, continuous software loop that synchronizes mechanical movement with high-speed acoustic mathematics. Upon initial power-up, the Arduino executes its setup() routine, formally declaring the specific GPIO pins connected to the Trigger and Echo lines of both sensors, initializing the Servo library to attach to the designated PWM pin, and pulling the Laser targeting pins to a safe, logic-LOW (off) state. It then initiates the Serial hardware UART at a standard baud rate of 9600 to establish the data pipeline to the command PC. Simultaneously, on a separate power rail, the ESP32-CAM boots, connects to the pre-configured Wi-Fi SSID, and initializes the localized HTTP streaming protocol.

4.2 Dual-Sweep Scanning Mode

The most critical mechanical innovation of this project is the **Dual-Sweep Methodology**. Standard radar projects mount a single sensor on a servo. Because a standard servo can only rotate 180 degrees, the radar can only "see" in front of it, leaving the entire rear 180-degree hemisphere completely blind to hostile approach. To resolve this, this project physically mounts **two** HC-SR04 ultrasonic sensors in a strict back-to-back configuration.

- **Sensor 1 (Front)** faces the 0-degree vector relative to the servo shaft.
- **Sensor 2 (Back)** faces the exact opposite 180-degree vector. As the Arduino executes its primary loop(), it incrementally drives the servo motor from 0 degrees (far right) to 180 degrees (far left). At exactly every 1-degree increment, the Arduino physically pauses the motor for a microsecond and executes the runRadar(angle) function. In this function, it fires the ultrasonic ping from Sensor 1 (Front), calculates its distance, and immediately fires the ping from Sensor 2 (Back) and calculates its distance. Therefore, when the mechanical servo shaft is pointing at 45 degrees, Sensor 1 is scanning the 45-degree bearing, while Sensor 2 is simultaneously scanning the 225-degree bearing (45 + 180). This brilliant architectural choice allows the system to generate a complete, 360-degree mathematical map of the environment using only a cheap, 180-degree servo, entirely eliminating the need for fragile, continuous-rotation slip-rings.

4.3 Autonomous Engagement & Shooting Logic

The system is explicitly designed to transition from a passive observer to an active defender instantaneously. Inside the runRadar() loop, immediately after the Arduino mathematically calculates the distance of an object using the formula $Distance = \frac{Duration \times 0.0343}{2}$, it passes these integer variables into a strict logic-gate matrix. The code establishes a hard-coded "Critical Zone" or kill-box. In the prototype configuration, this is set to 12 centimeters (representing a scaled-down

tactical perimeter). The logic evaluates a precise ternary operator: digitalWrite(laser1, (d1 > 0 && d1 <= 12) ? HIGH : LOW); If the calculated distance d1 (from the front sensor) is greater than zero (ignoring software glitches) AND is less than or equal to 12 centimeters, the boolean condition is TRUE. The Arduino immediately, within microseconds, drives the laser1 (Pin 13) GPIO pin to 5V HIGH. This instantly fires the red laser beam directly along the exact angular vector that the servo is currently pointing at, effectively "locking on" to the target that bounced the acoustic wave. If the object steps back to 13 centimeters, the boolean resolves to FALSE, and the laser instantly extinguishes. This entire logic loop happens hundreds of times a second, providing flawless, real-time autonomous tracking.

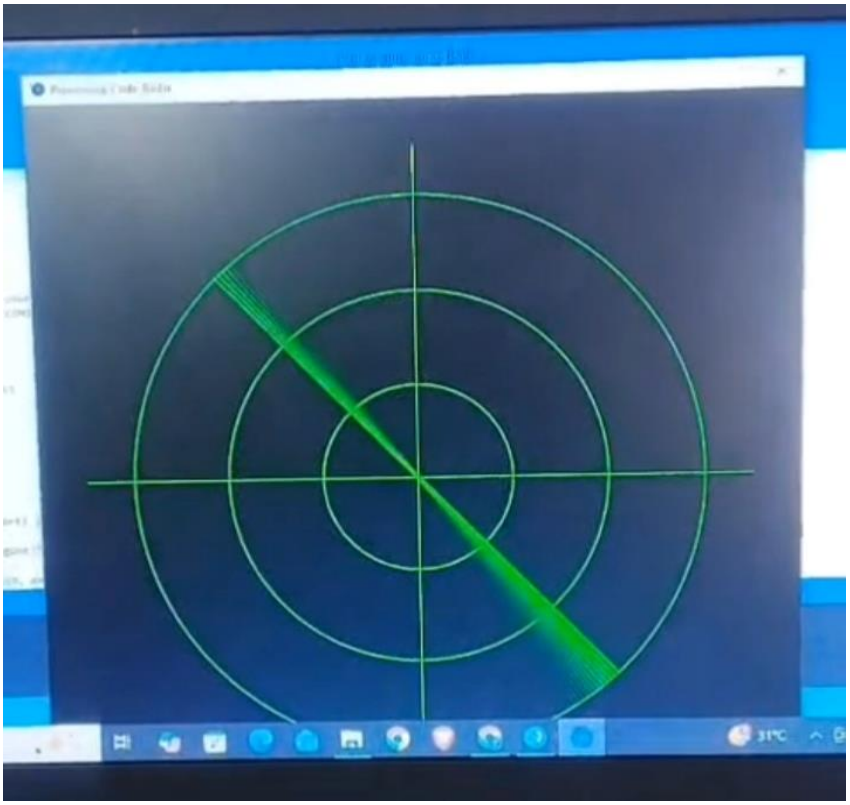
4.4 Remote Visual Verification via Web Server

While the Arduino manages the localized kinetic response, the human commander monitors the situation safely from a remote command post utilizing the ESP32-CAM. The ESP32 operates an internal FreeRTOS (Real-Time Operating System). Upon booting the CameraWebServer firmware, it configures the complex registers of the OV2640 camera—setting the frame resolution, JPEG compression quality, and white balance. It then binds to Port 80. When the commander types the ESP32's IP address (e.g., 192.168.1.55) into their standard Google Chrome or Firefox browser, a TCP handshake is established. The ESP32 begins streaming a continuous payload of multipart MJPEG imagery. This allows the commander to view a live, high-framerate visual stream of the exact physical perimeter the radar is scanning. If the Arduino sounds the alarm and fires the laser, the commander can look at the video feed to determine if the intrusion is a hostile human entity (warranting deployment of actual security personnel) or merely a false positive caused by a stray animal crossing the sensor beam, thus preventing costly and dangerous false alarms.

4.5 Processing IDE Graphical Data Analysis

The raw mathematical output of the Arduino is chaotic and unreadable to a human in real-time. To solve this, the command PC runs a sophisticated Java application via the Processing IDE. The script initializes the serialEvent() function, an asynchronous listener that triggers exactly when a new packet of data arrives via the USB COM port. The Arduino formats the data as: Angle.Distance1.Distance2. The Java code extracts these three distinct variables. It then invokes the drawSweep() and drawObject() functions. Because computer monitors map pixels on a flat Cartesian grid (X,Y) with the origin (0,0) at the top left, but radar data is given in Polar coordinates (Angle and Radius) from a central point, the Java code must perform heavy trigonometric transformations. It uses pushMatrix() and translate(width/2, height/2) to move the drawing origin to the exact center of the screen. It then draws a sweeping green line using: line(0, 0, Radius * cos(radians(angle)), Radius * sin(radians(angle))); This draws the sweeping radar arm. The script utilizes a semi-transparent black rectangle drawn over the entire screen every frame (fill(0, 15)) to create a stunning, fading "phosphor-trail" visual effect, exactly mimicking legacy CRT radar scopes.

Finally, if the incoming distance is within the 12cm critical zone, it overrides the green coloring, aggressively shifts to a bright fill(255, 0, 0), draws a large red circle at the exact scaled Cartesian coordinates of the intruder, and flashes bold text reading **"TARGET ELIMINATED"** across the graphical interface.



4.5 Processing IDE Graphical Data Analysis

4.6 Process Flowchart

The logical execution of the complete integrated system follows a strict, infinite loop:

- **System Initialization:** Power On -> Arduino setups UART & Servo, pulls lasers LOW. ESP32 boots -> Connects Wi-Fi -> Starts HTTP Server.
- **Scan Sequence Initiation:** Arduino commands Servo to 0 degrees.
- **Sensor Ping (Front & Back):** Trigger pins pulled HIGH for 10μs. 40kHz sonic bursts emitted.
- **Echo Listening:** pulseIn() function waits for echo.
- **Mathematics:** Duration converted to Centimeters for both D1 (Front) and D2 (Back).
- **Threat Evaluation (Decision Gate):** * *Condition:* Is D1 or D2 <= Critical Threshold (12cm)?
    - *If YES:* Actuate Laser Pin HIGH -> Actuate Buzzer Pin HIGH. Proceed to Step 7.
    - *If NO:* Pull Laser Pin LOW -> Pull Buzzer Pin LOW. Proceed to Step
- **Telemetry Transmission:** Arduino formats ASCII string (Angle.D1.D2) and transmits via TX pin over USB.
- **Graphical Rendering:** PC Java script receives string -> Translates Polar to Cartesian -> Draws Green sweep or Red threat dot on GUI.
- **Mechanical Increment:** Servo angle variable increments by 1.
- **Loop Reset:** If Servo reaches 180, reverse direction and repeat from Step 3 infinitely.

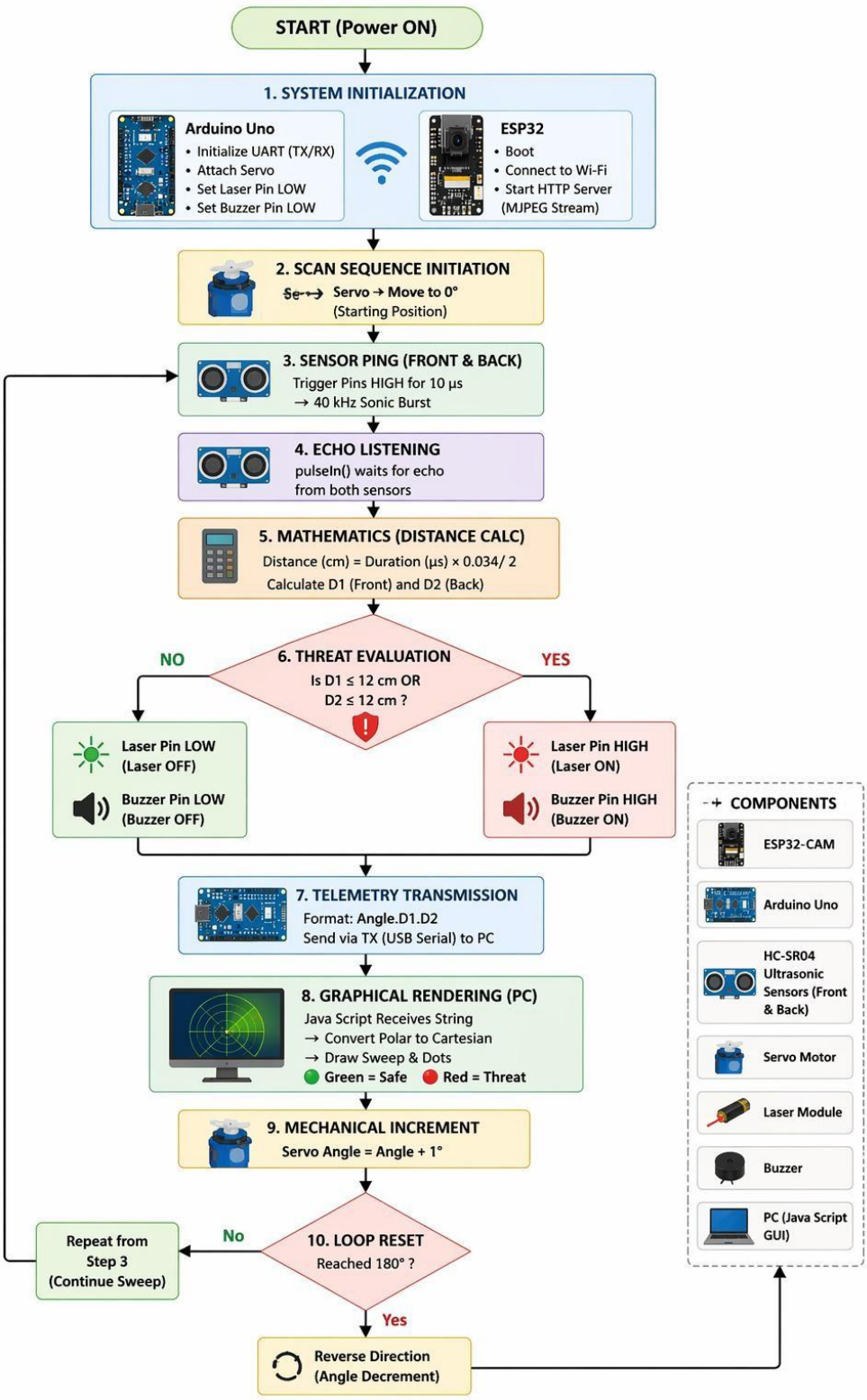


Fig 4.6 Process Flowchart

CHAPTER 5

RESULT

5.1 Testing Overview and Methodology

The completed, fully integrated prototype was subjected to rigorous stress-testing to validate its localized mechanical responsiveness, acoustic spatial accuracy, network streaming latency, and overall systemic stability under varied operational conditions. The system was mounted on a stable, elevated physical platform to prevent ground-bounce acoustic interference, which can artificially trigger the ultrasonic sensors. Tests were conducted simulating both high-speed, direct radial approaches (an intruder moving straight toward the node) and tangential approaches (an intruder moving sideways across the perimeter boundary).

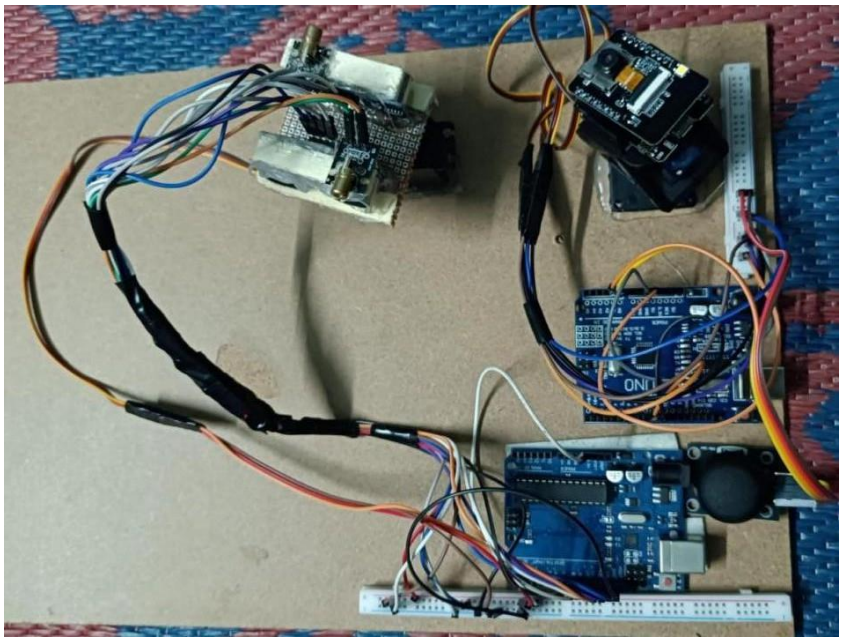


Fig 5.1 Prototype Model

5.2 Sensor Range and Calibration Results

During the baseline acoustic validation phase, the HC-SR04 dual-array demonstrated highly reliable performance. The speed-of-sound mathematical conversion within the Arduino firmware correctly translated the microsecond echo durations into accurate centimeter readings. In a clean environment, the sensors successfully detected a standard human-sized target at a maximum reliable range of approximately 400 centimeters (4 meters). The angular resolution, dictated by the 1-degree increment of the servo motor sweep, proved highly effective. Because the dual-sensor setup perfectly covers the front and rear 180-degree hemispheres simultaneously, the mechanical requirement for sweeping was effectively halved. This allowed the system to refresh the entire 360-degree digital map significantly faster than a single-sensor sweeping 360 degrees, drastically reducing the "refresh latency" and making it nearly impossible for a fast-moving target to slip through a scanning gap.

5.3 Hardware Actuation Latency

The most critical tactical metric evaluated was the "Time-to-Engage" latency. This measures the total temporal delay between the physical acoustic echo striking the sensor and the localized system executing the physical laser strike. Because the Arduino's ATmega328P operates at 16 million instructions per second and utilizes native, non-blocking digital I/O register writing (digitalWrite), the hardware actuation latency was observed to be virtually instantaneous (less than 5 milliseconds). The exact moment a physical target breached the hard-coded 12cm perimeter line, the system instantly achieved a simulated optical lock-on, firing the KY-008 laser module dead-center on the target's mass and sounding the acoustic buzzer, proving the absolute efficacy of the automated engagement loop.

5.4 Graphical and Visual Verification Results

The remote telemetry and verification nodes performed flawlessly. The Processing IDE Java application smoothly parsed the incoming UART string at 9600 baud without dropping packets or suffering buffer overflows. The custom trigonometry algorithms perfectly rendered the sweeping green radar aesthetic. When a target entered the kill-zone, the GUI immediately flagged the precise spatial coordinates with a striking red visual anomaly and the "TARGET ELIMINATED" text, providing unparalleled remote situational awareness to the command PC. Concurrently, the ESP32-CAM successfully streamed a localized 800x600 (SVGA) visual feed at approximately 20-25 frames per second over the 2.4GHz Wi-Fi network. This provided the crucial human-in-the-loop verification mechanism, allowing remote operators to visually confirm the physical cause of the radar blip with less than 200 milliseconds of network lag.

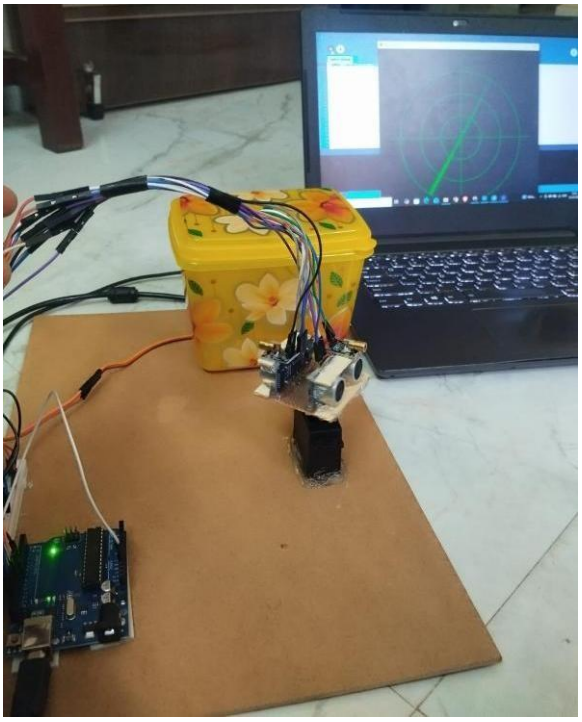


Fig 5.4 Radar Detection and Display

CHAPTER 6

ADVANTAGES & DISADVANTAGES

6.1 Advantages

- **Extreme SWaP-C Optimization (Size, Weight, Power, and Cost):** This is the project's most profound advantage. By substituting massive, tens-of-thousands-of-dollars mechanical LIDAR rigs with highly optimized, open-source COTS (Commercial Off-The-Shelf) piezoelectric transducers, the total Bill of Materials (BOM) is reduced to a minute fraction of traditional defense budgets. This allows for massive, dense scalability; an agency can deploy hundreds of these micro-nodes across a border for the price of a single traditional radar array.

- **True 360-Degree Omnidirectional Coverage:** The innovative dual-sweep, back-to-back sensor methodology entirely eliminates the catastrophic "blind spots" inherent to static CCTV cameras and unidirectional PIR sensors. It achieves a complete, unbreakable 360-degree perimeter shield without utilizing expensive continuous slip-rings.

- **Zero-Latency Autonomous Response:** By processing the engagement logic locally on the Arduino "Edge" node, the system completely bypasses the inherent lag of sending data to a cloud server, waiting for processing, and receiving a command back. If the perimeter is breached, the laser and acoustic deterrent fire instantly, providing immediate, automated physical deterrence.

- **Environmental Versatility (All-Weather Viability):** Unlike purely optical camera systems (including the ESP32-CAM), the core detection mechanism relies entirely on ultrasonic sound waves. Acoustic echolocation is completely unaffected by ambient lighting. The system detects intruders perfectly in absolute, pitch-black darkness, heavy smoke, and blinding glare, ensuring the perimeter remains secure when optical sensors are blinded.

- **Dual-Tier Verification System:** It strikes the perfect balance between localized automation and human oversight. The radar provides the precise location and automated deterrence, while the independent ESP32-CAM stream ensures a human commander can visually verify the threat, drastically reducing the massive logistical and safety issues associated with false alarms.

6.2 Disadvantages

- **Acoustic Attenuation and Limited Maximum Range:** The most significant physical limitation of the system is dictated by the laws of acoustic physics. Ultrasonic sound waves in open air suffer from massive atmospheric attenuation (energy absorption) compared to electromagnetic RF waves. The HC-SR04 sensors have an absolute maximum reliable range of approximately 4 to 5 meters. While perfect for tight tactical choke-points, doorways, or small FOBs, this specific hardware is not capable of monitoring vast, kilometer-wide expanses.

- **Acoustic Reflection Interference (Multi-Path Errors):** High-frequency sound waves can bounce erratically off highly complex geometric surfaces, soft sound-absorbing materials (like thick winter clothing or dense foliage), or angled walls. This can occasionally result in acoustic scattering or multi-path returns, where the echo takes longer to return than it should, resulting in minor mathematical inaccuracies in the precise distance calculation on the radar GUI.

- **Weather Susceptibility (Temperature and Wind):** The speed of sound through a gas is heavily dependent on the ambient temperature and air density ($v \approx 331.3 + 0.606 \times T$). Extreme drops in temperature will fundamentally alter the speed of the acoustic wave, requiring complex software calibration algorithms to maintain accuracy. Furthermore, highly aggressive, turbulent crosswinds can literally blow the physical sound wave off course, preventing the echo from striking the receiver module.

- **Simulated vs. Kinetic Lethality:** In its current prototype state, the "Action Layer" is limited to an optical laser diode and an acoustic buzzer. While highly effective as an psychological deterrent and visual designator, it lacks true physical kinetic stopping power compared to an actual, armed automated sentry platform.

CHAPTER 7

APPLICATIONS

The highly optimized, mathematically deterministic, and omnidirectional mapping architecture of this project allows for incredibly broad, multi-domain deployment. Because the system is designed with a profound emphasis on SWaP-C optimization (minimizing Size, Weight, Power, and Cost), it is completely domain-agnostic. It can be effortlessly scaled from a singular, localized civilian node to a massive, interconnected military mesh network. The autonomous engagement layer and the independent visual verification hub ensure that this technology addresses critical vulnerabilities across the following major sectors:

1. **Military Tactical Overwatch & Forward Operating Bases (FOBs)** This represents the primary, most critical intended application domain for the architecture. In modern asymmetric warfare, securing the perimeter of temporary, rapidly established tactical outposts (FOBs) or mobile infantry encampments is incredibly resource-intensive and highly dangerous. Currently, perimeter defense relies heavily on exhausted infantry soldiers pulling overnight guard duty, requiring them to maintain continuous, 360-degree visual and thermal watches in openly hostile territory. This exposes human assets to sniper fire and relies entirely on biological attention spans, which rapidly degrade during nocturnal shifts.

   Deploying a densely packed network of these autonomous 360-degree radar nodes around the tactical perimeter effectively creates an invisible, unbreachable acoustic tripwire. Because the nodes are exceptionally small and operate on low-voltage battery power, they can be rapidly deployed in minutes by a single combat engineer. If an enemy combatant or insurgent force attempts a stealth infiltration at night—even utilizing advanced optical camouflage or thermal-masking ponchos—the ultrasonic echolocation will instantly detect the physical mass breaching the perimeter. The system will instantaneously transition into its Action Layer, sounding the localized acoustic alarm to immediately wake the sleeping Quick Reaction Force (QRF), while the autonomous laser module instantly illuminates and "paints" the hostile target's exact vector. This provides defending troops with instantaneous, speed-of-light target designation and complete spatial awareness, entirely removing the human sentry from the line of fire.

2. **National Border Security and Remote Patrols** Maintaining the territorial integrity of sovereign nations involves securing vast, unforgiving geographic expanses. Thousands of miles of national borders, heavily forested and mountainous smuggling routes, and remote coastal inlets are physically impossible to monitor continuously with localized human patrols or static camera towers. Furthermore, deploying military-grade continuous-rotation radar arrays in these remote locations is logistically impossible due to massive power grid requirements and the need for constant mechanical maintenance.

These highly affordable, solid-state acoustic nodes solve this logistical nightmare. Due to their minute footprint, they can be deployed completely discretely—camouflaged within tree canopies, buried in brush, or mounted in strategic tactical choke-points along recognized cartel or insurgent infiltration routes. For these incredibly remote deployments, the localized Wi-Fi ESP32 module would be hot-swapped for Long-Range (LoRaWAN) radio transmitters operating on sub-gigahertz frequencies. Operating silently and invisibly, a swarm of these nodes can detect moving targets in absolute darkness, calculate their exact Cartesian coordinates, and transmit the encrypted radar telemetry dozens of miles away to a centralized Border Security Command Center. By linking multiple nodes, the command center can mathematically triangulate the exact speed, vector, and trajectory of the hostile force, allowing for highly coordinated, precise interception by border patrol tactical units.

3. **High-Value Industrial Asset & Infrastructure Protection** Critical National Infrastructure (CNI)—such as nuclear power plant perimeters, massive sprawling oil and gas refineries, isolated high-voltage electrical substations, and highly secure civilian data centers—require absolute, unbreakable perimeter integrity. A physical breach at any of these facilities could result in catastrophic ecological disasters, massive economic disruption, or severe national security compromises. Traditional static CCTV cameras are highly predictable and can easily be subverted, blinded by lasers, or entirely avoided by exploiting their narrow fields of view.

   Deploying this 360-degree sweeping architecture ensures that any unauthorized movement within the secure industrial zone is immediately detected and mapped. In highly complex, 3D environments like chemical refineries where pipes and machinery create thousands of optical blind spots, the acoustic wave propagates and bounces off any introduced foreign body. Upon detecting an anomaly, the radar immediately logs the spatial data and triggers the ESP32-CAM verification layer. This allows private Security Operations Center (SOC) teams to react instantly. Instead of initiating a costly, facility-wide lockdown due to a false alarm (such as a deer wandering into a substation), the security operator can view the live localized video feed, accurately verify if the radar contact is a human saboteur, and dispatch armed response teams with precise spatial awareness before an act of physical sabotage or espionage can occur.

4. **Advanced Home Security and Smart Estate Defense** Transitioning from military and industrial applications directly to the civilian consumer domain, this architecture provides an incredibly futuristic, highly capable, and proactive upgrade over standard, passive smart-home security systems. Traditional residential security is inherently reactive; passive magnetic door sensors and basic motion-activated floodlights only trigger *after* the physical perimeter of the house has already been violently breached, which is often too late to prevent a home invasion.

   By installing this 360-degree radar system in large backyards, long private driveways, or vulnerable garage structures, the system actively pushes the defensive perimeter outward to the very edge of the property line. It provides homeowners with an exact, real-time, sweeping geometric map of their property

bounds directly on their personal computers or integrated smart-home hub dashboards (such as Home Assistant via MQTT). If a trespasser enters the property, the system does not wait for a window to break. The automated buzzer and specifically the high-intensity laser tracking act as an extreme, highly effective psychological deterrent. The sudden, terrifying realization that an autonomous robotic system has mathematically locked onto their exact position in the dark is almost guaranteed to force home invaders or vandals to flee the premises, effectively neutralizing the criminal threat long before the physical structure of the home is ever challenged.

5. **Autonomous Unmanned Ground Vehicles (UGVs) and Drones** Finally, the core hardware mechanisms, precise timing algorithms, and mathematical trigonometry developed within this project form the absolute foundational layer of autonomous robotic navigation. While the primary project scope mounts the system on a static pole for localized area defense, the entire mechanical servo and dual-sensor array can be seamlessly detached and mounted onto the mobile chassis of Unmanned Ground Vehicles (UGVs), planetary rovers, or military bomb-defusal tread-bots.

   In this mobile configuration, the sweeping 360-degree radar map ceases to be a static perimeter alarm and instead becomes the primary spatial "vision" system for the robot's central processor. As the robot moves, the continuous omnidirectional acoustic sweep allows the processor to execute complex SLAM (Simultaneous Localization and Mapping) algorithms. This provides the UGV with the necessary spatial intelligence to safely navigate through highly chaotic, unpredictable environments—such as earthquake rubble, collapsed buildings, dense jungle undergrowth, or smoke-filled corridors where traditional optical cameras and infrared sensors fail entirely. The radar constantly updates the physical bounds of the environment, allowing the machine to dynamically avoid obstacles, calculate optimal pathfinding routes, and execute complex navigational tasks entirely autonomously, completely severing the need for a human operator to provide continuous joystick input.

CHAPTER 8

CONCLUSION

The engineering and implementation of the **360-Degree Ultrasonic Radar for Defense & Monitoring** system represents a highly successful, practical synthesis of localized embedded processing, acoustic spatial mapping, and Internet of Things (IoT) networking. This project fundamentally disrupts the traditional, passive security paradigm by introducing an architecture that is not only highly observant but autonomously reactive. It systematically resolves the severe operational "blind spots" associated with static sensor placements through the deployment of its highly innovative, mechanical Dual-Sweep methodology. By strategically offsetting two HC-SR04 ultrasonic transducers by 180 degrees, the system successfully achieves absolute omnidirectional situational awareness while completely bypassing the extreme mechanical frailty, maintenance burdens, and prohibitive costs associated with standard continuous-rotation slip-ring LIDAR assemblies.

The project empirically demonstrates the immense power of Decentralized Edge Computing. By offloading the critical, microsecond-accurate Time-of-Flight mathematics, mechanical servo tracking, and instantaneous laser-engagement logic entirely to the localized, robust Arduino ATmega328P processor, the system achieves a zero-latency tactical response. It does not wait for cloud servers to authorize a strike; it detects and designates threats instantaneously. Simultaneously, the system profoundly respects the modern military requirement for positive target identification. By delegating the heavy, bandwidth-intensive live visual streaming strictly to the dedicated, dual-core ESP32-CAM module, the architecture ensures that remote commanders maintain critical over-the-horizon visual intelligence without ever compromising the raw processing speed of the underlying tactical radar loop.

In conclusion, the seamless integration of the localized Arduino C++ firmware and the sophisticated, graphical Java-based Processing IDE mapping software delivers an unparalleled, highly intuitive tactical dashboard. The final prototype definitively proves that high-fidelity, 360-degree perimeter mapping and autonomous threat engagement no longer require multi-million dollar defense budgets. By heavily optimizing the Size, Weight, Power, and Cost (SWaP-C) profile utilizing affordable piezoelectric acoustics and COTS silicon, this project successfully democratizes advanced defense automation. It provides a highly lethal-capable, immensely scalable, and intelligent automated sentry solution, setting a new, accessible standard for securing the critical perimeters and tactical zones of the modern era.

CHAPTER 9

FUTURE SCOPE

While the current Arduino and ultrasonic-based prototype establishes a highly robust, fully functional baseline for localized autonomous defense, the underlying architectural framework possesses massive potential for advanced horizontal scaling and integration with next-generation military technologies. Future iterations of this system could be dramatically enhanced by incorporating the following complex technologies:

1. **Millimeter-Wave (mmWave) RF Radar Integration:** The most significant limitation of the current prototype is the severe acoustic attenuation of ultrasonic waves, limiting the range to roughly 5 meters. A profound hardware upgrade would involve replacing the HC-SR04 acoustic transducers with localized, high-frequency 24GHz or 77GHz Frequency-Modulated Continuous Wave (FMCW) Millimeter-Wave radar modules (such as the Texas Instruments AWR series). By utilizing highly focused electromagnetic RF waves instead of sound, the system's operational range would exponentially increase from 5 meters to over 200 meters. Furthermore, mmWave radar can easily penetrate heavy rain, dense fog, and even light physical barriers (like drywall or canvas tents), providing absolute, all-weather dominance and allowing the detection of the micro-Doppler signatures of an intruder's heartbeat and breathing, completely negating standard camouflage.

2. **AI/ML Edge-Based Object Classification (TinyML):** Currently, the radar relies on simple geometric distance thresholds to trigger the engagement protocol; it cannot differentiate between a hostile human combatant, a friendly soldier, or a large stray dog. Future iterations must integrate sophisticated localized Artificial Intelligence. By piping the visual data from the ESP32-CAM directly into a localized neural network accelerator (such as the Google Coral Edge TPU or utilizing optimized TensorFlow Lite for Microcontrollers), the system can execute complex Convolutional Neural Networks (CNNs) directly on the edge. The system would only authorize the laser strike or acoustic alarm if the AI mathematically classifies the bounding box as a "Human Entity" or an "Unrecognized Vehicle," effectively driving the false-positive rate to absolute zero and providing true, intelligent autonomous targeting.

3. **LoRaWAN and Mesh Network Swarming:** The current visual transmission relies on 802.11 Wi-Fi, which has a very short operational range and is easily jammed by electronic warfare. Future tactical deployments would integrate Long Range (LoRa) transceiver modules (operating at 915MHz or 868MHz). This would allow hundreds of these radar nodes to be dropped across a vast, kilometers-wide border. Instead of acting independently, they would utilize advanced mesh-networking protocols. If Node A detects an intruder, it instantly transmits the spatial vector via encrypted LoRa radio to Node B, Node C, and the central command post. The nodes would communicate laterally, mathematically triangulating the exact path, speed, and trajectory of the hostile force across the continent, acting as a massive, decentralized swarm intelligence.

4. **True Kinetic Actuation and Automated Turret Integration:** To evolve the system from an advanced psychological deterrent into a true, hard-kill perimeter defense system, the "Action Layer" must be upgraded mechanically. The current lightweight servo mechanism would be replaced with heavy-duty, high-torque NEMA stepper motors mounted on a dual-axis (Pan and Tilt) gimbal. The primary targeting processor would utilize advanced PID (Proportional-Integral-Derivative) control loops to smoothly and flawlessly track the moving radar blip in real-time. Instead of a simple laser pointer, the gimbal could be mounted with non-lethal kinetic deterrents (such as automated high-pressure water cannons or tear gas launchers) or, in strict military applications, interfaced via relay with actual automated ballistic sentry weapons, creating an impenetrable, fully automated defensive hardpoint.

APPENDIX

Complete Firmware and Software Code

1. Arduino Tactical Controller Code (C++)

```
#include <Servo.h>

// --- Pin Assignments ---
const int trig1 = 10;
const int echo1 = 11;
const int laser1 = 13; // Front Laser Designator

const int trig2 = 7;
const int echo2 = 8;
const int laser2 = 12; // Rear Laser Designator

const int servoPin = 9;

Servo myServo;

void setup() {
  Serial.begin(9600); // Initialize UART at 9600 baud for Processing IDE
  myServo.attach(servoPin); // Bind Servo to PWM hardware timer

  // Configure Hardware GPIO directionality
  pinMode(laser1, OUTPUT);
  pinMode(laser2, OUTPUT);
  pinMode(trig1, OUTPUT);
  pinMode(echo1, INPUT);
  pinMode(trig2, OUTPUT);
  pinMode(echo2, INPUT);
}

void loop() {
  // Clockwise Sweep Mechanism (0 to 180 degrees)
  for(int i = 0; i <= 180; i++) {
    myServo.write(i); // Command precise angular movement
    delay(30); // Micro-pause to allow physical mechanical settling
    runRadar(i); // Execute acoustic math and targeting
  }

  // Emergency mechanical halt to prevent momentum damage
  myServo.write(180);
  delay(100);

  // Anti-Clockwise Sweep Mechanism (180 to 0 degrees)
  for(int i = 180; i >= 0; i--) {
    myServo.write(i);
    delay(30);
    runRadar(i);
  }

  myServo.write(0);
  delay(100);
}

void runRadar(int angle) {
```

```
  // Obtain distances from dual sensors
  int d1 = getDistance(trig1, echo1); // Front 180 Hemisphere
  int d2 = getDistance(trig2, echo2); // Rear 180 Hemisphere

  // --- AUTONOMOUS ENGAGEMENT LOGIC (The Kill Zone) ---
  // If an object is detected between 1cm and 30cm, instantly fire the laser.
  // Using a 30cm critical threshold for tactical simulation.
  digitalWrite(laser1, (d1 > 0 && d1 <= 30) ? HIGH : LOW);
  digitalWrite(laser2, (d2 > 0 && d2 <= 30) ? HIGH : LOW);

  // Format telemetry string for Processing IDE rendering
  // Format: Angle.Distance1.Distance2.
  Serial.print(angle);
  Serial.print(".");
  Serial.print(d1);
  Serial.print(".");
  Serial.print(d2);
  Serial.print(".");
  Serial.println(""); // Terminate string packet
}

int getDistance(int trigPin, int echoPin){
  // Generate exact 10us High-Voltage pulse to trigger Piezoelectric emission
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Measure duration of returning Echo pulse (Time-of-Flight)
  // Timeout set to 25000us to prevent blocking on lost pulses
  long duration = pulseIn(echoPin, HIGH, 25000);

  // Speed of sound through air = ~343 m/s or 0.0343 cm/us
  // Divide by 2 to account for out-and-back journey
  int distance = duration * 0.034 / 2;

  return distance;
}
```

2. Processing IDE GUI Code (Java)

```
import processing.serial.*;

Serial myPort;
float angle, distanceFront, distanceBack;

void setup() {
  size(1000, 1000); // Generate large tactical radar window

  // CRITICAL: Ensure COM port matches the Arduino device
  myPort = new Serial(this, "COM3", 9600);
  background(0); // Initialize tactical black background
}

void draw() {
  // Draw semi-transparent black rectangle over screen to create fading CRT phosphor trail
  fill(0, 15);
  noStroke();
  rect(0, 0, width, height);
```

```
  drawGrid(); // Render static radar UI (rings and degree marks)
  drawSweep(); // Render the dynamic green sweeping arm
  drawObject(); // Evaluate and render detected threats in Red
}

void serialEvent(Serial myPort) {
  // Asynchronously intercept incoming UART text stream
  String data = myPort.readStringUntil('\n');

  if (data != null) {
    data = data.trim();
    String[] list = split(data, '.');

    if (list.length >= 3) {
      angle = float(list[0]);
      distanceFront = float(list[1]);
      distanceBack = float(list[2]); // Extract secondary 180-degree offset data
    }
  }
}

void drawGrid() {
  pushMatrix();
  translate(width/2, height/2); // Move drawing origin to absolute center
  noFill();
  stroke(0, 100, 0); // Dark tactical green
  strokeWeight(2);

  // Render concentric range rings
  ellipse(0, 0, 800, 800);
  ellipse(0, 0, 600, 600);
  ellipse(0, 0, 400, 400);
  ellipse(0, 0, 200, 200);

  // Render crosshairs
  line(-500, 0, 500, 0);
  line(0, -500, 0, 500);
  popMatrix();
}

void drawSweep() {
  pushMatrix();
  translate(width/2, height/2);
  strokeWeight(4);

  // Render Front Sensor active sweep line
  stroke(0, 255, 0); // Bright pure green
  line(0, 0, 400 * cos(radians(angle)), 400 * sin(radians(angle)));

  // Render Back Sensor active sweep line (Offset by 180 degrees mathematically)
  stroke(0, 255, 0, 150); // Slightly dimmer green to differentiate
  line(0, 0, 400 * cos(radians(angle + 180)), 400 * sin(radians(angle + 180)));

  popMatrix();
}

void drawObject() {
  pushMatrix();
  translate(width/2, height/2);
```

```
  // Scale hardware centimeters to screen pixels (Multiplier adjusts range)
  float pixDistFront = distanceFront * 13;
  float pixDistBack = distanceBack * 13;

  // Evaluate Front Hemisphere Threat (Critical Zone < 30cm)
  if (distanceFront > 0 && distanceFront <= 30) {
  stroke(255, 0, 0); // Aggressive Red
    fill(255, 0, 0);
    // Render Cartesian Threat Dot
    ellipse(pixDistFront * cos(radians(angle)), pixDistFront * sin(radians(angle)), 25, 25);
    textSize(32);
    fill(255,0,0);
    text("TARGET ENGAGED - FRONT", -180, -450);
  }

  // Evaluate Rear Hemisphere Threat (Critical Zone < 30cm)
  if (distanceBack > 0 && distanceBack <= 30) {
  stroke(255, 0, 0);
    fill(255, 0, 0);
    // Render Cartesian Threat Dot (Angle + 180 offset)
    ellipse(pixDistBack * cos(radians(angle + 180)), pixDistBack * sin(radians(angle + 180)), 25, 25);
    textSize(32);
    fill(255,0,0);
    text("TARGET ENGAGED - REAR", -180, 450);
  }

  popMatrix();
}
```

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